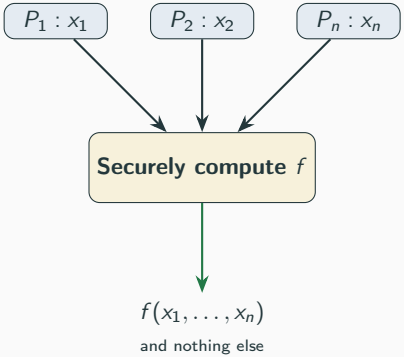
Non-Interactive MPC, (Quantumly) Revisited

Prabhanjan Ananth[1] Divyanshu Bhardwaj[1] Aparna Gupte[2]

[1]UC Santa Barbara [2]MIT

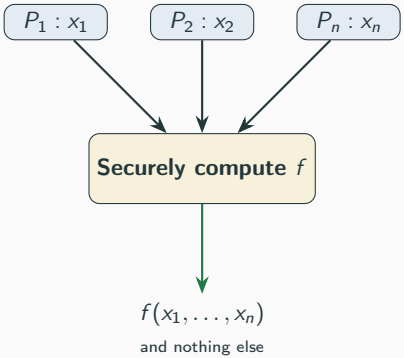
[1] ePrint: 2026/302

Secure Multi-Party Computation (MPC)



A foundational cryptographic goal studied since the late 1980s [GMW87, BGW88, CCD88].

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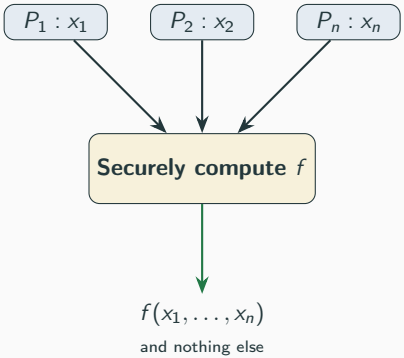


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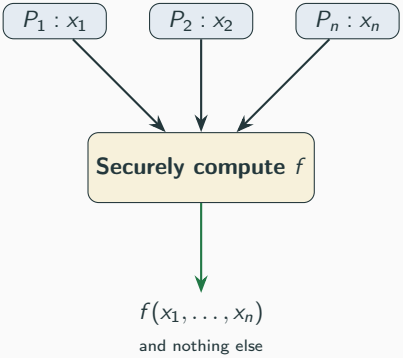
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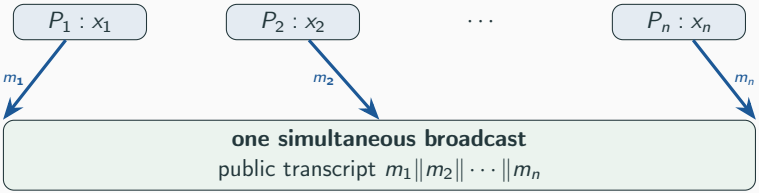
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Malicious security

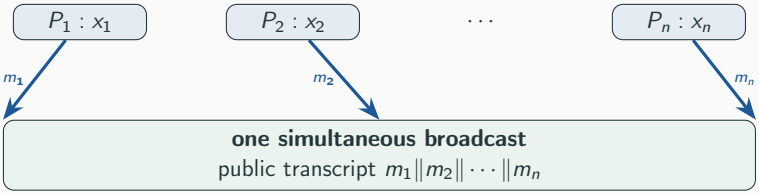
Corrupted parties may deviate arbitrarily; security must still hold.

Dream Goal: One-Round MPC



Each P_i chooses m_i without seeing any other party's message.

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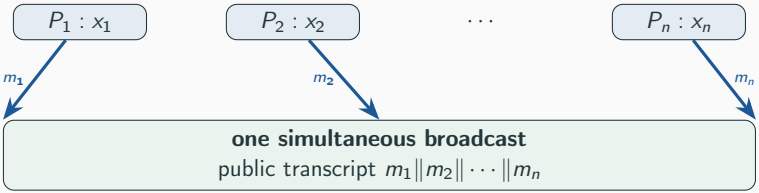


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Can security come with no additional interaction?

Why One-Message Classical MPC Fails: Resetting Attacks



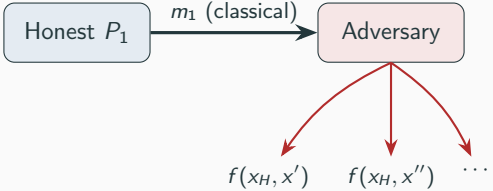
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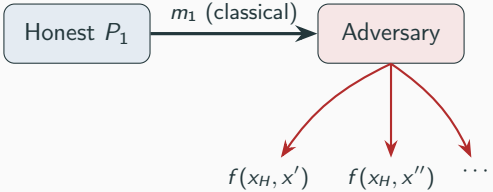
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copy & re-evaluate at will

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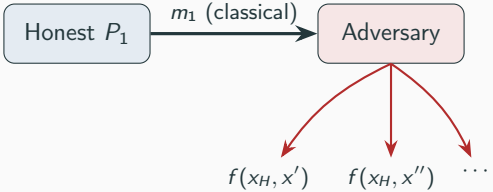
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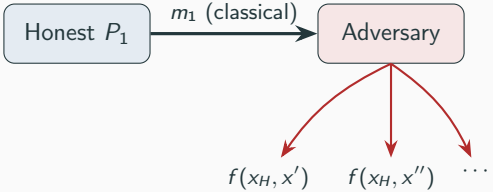
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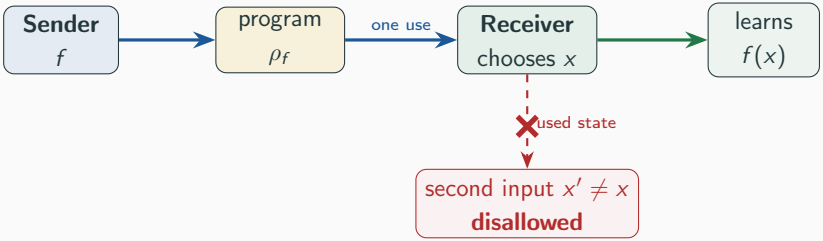
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Can we avoid leaking the entire residual function?

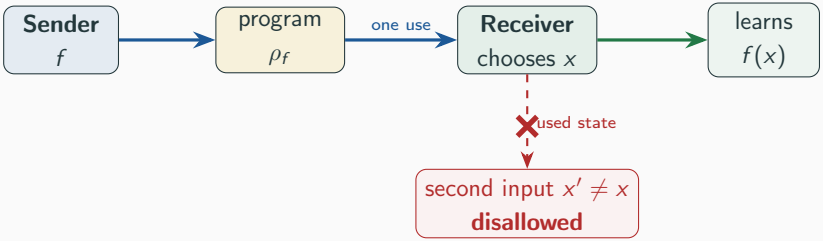
Quantum One-Time Programs



One-time guarantee

The receiver chooses *one* input and learns one output; the program cannot be reset for a second input.

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Starting point [GLR+25]

Quantum one-time programs are equivalent to semi-honest one-message 2PC (oracle model; plain-model results via cryptography).

The Ideal MPC Security: One Residual-Function Query

Fix the honest inputs x_H and define the residual function

$$f^H(\cdot) = f(x_H, \cdot).$$

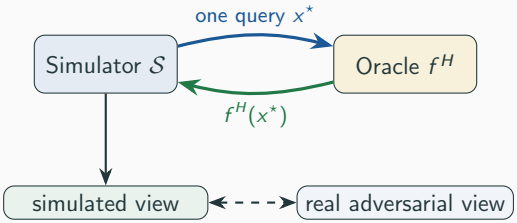
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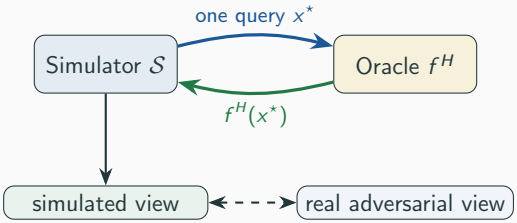
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$$\mathrm{View}_{\mathrm{real}}(\mathcal{A}) \approx \mathrm{View}_{\mathrm{ideal}}(\mathcal{S}^{f^H}).$$

Why One Physical Quantum Query Is Too Weak

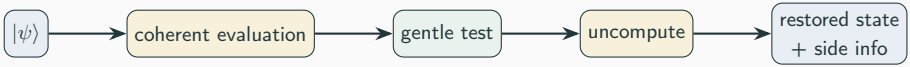
Barrier, even for two parties [GKR08, BGS13, GLR+25]

A real adversary can evaluate the honest procedure *coherently*, perform a gentle measurement, and uncompute. A simulator limited to one standard physical query to f^H cannot reproduce all of that side information.

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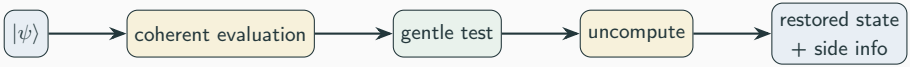
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What we need

A relaxation that permits coherent query/unquery behavior while still bounding how many *effective evaluations* can leak.

Oracle Model: Single-Effective-Query (SEQ) Simulation [GLR+25]

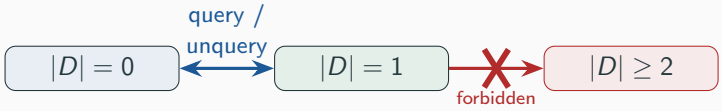
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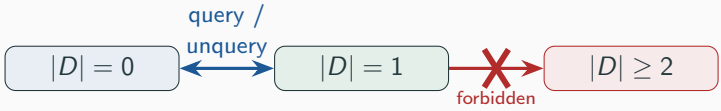
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Leakage guarantee

Even with polynomially many physical calls, an adversary extracts information corresponding to at most **one effective evaluation** of f^H.

This Work: Pre-Processing Determines the Leakage

Quantum pre-processing

1 SEQ oracle

Leakage: at most one output of f^H.

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Classical pre-processing

c SEQ oracles

With c corruptions, leakage is at most c outputs of f^H.

Main Tool: Coset States and Direct Product Hardness

Hidden subspace $A \subseteq \mathbb{F}_2^\lambda$, $\dim A = \lambda/2$ and offsets s, s':

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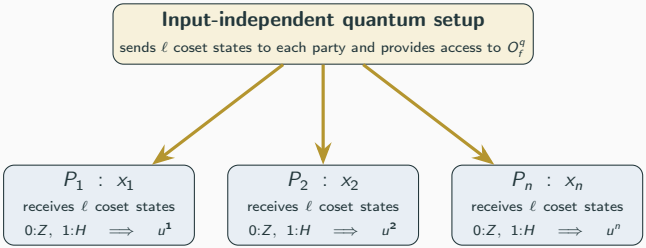
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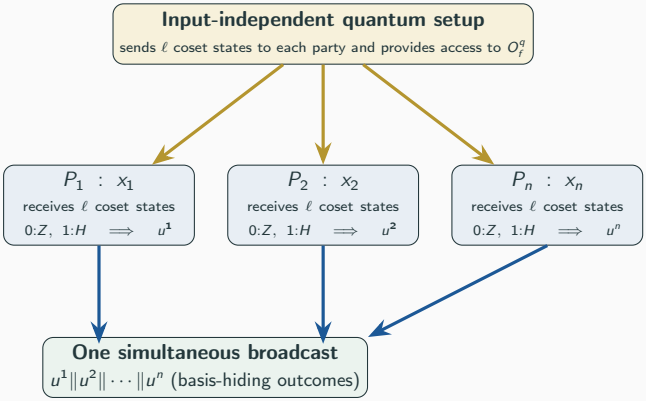
Direct product hardness (DPH) [CLLZ21, CHV23]

Given *one copy*, no adversary can produce two distinct valid vectors (two in $A + s$, two in $A^\perp + s'$, or one in each).

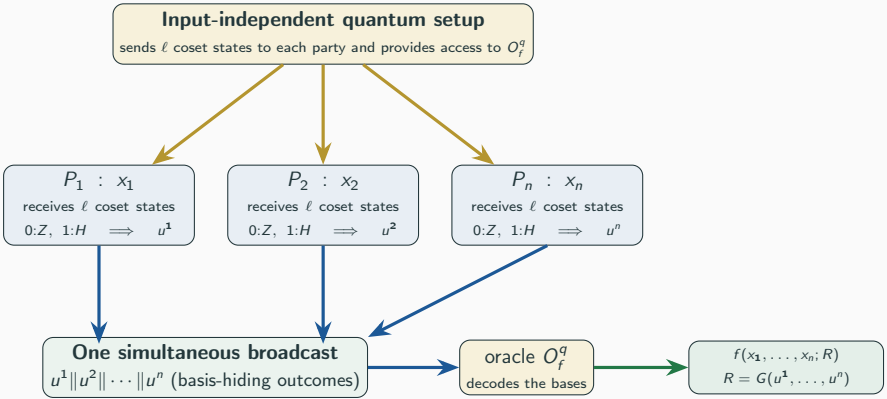
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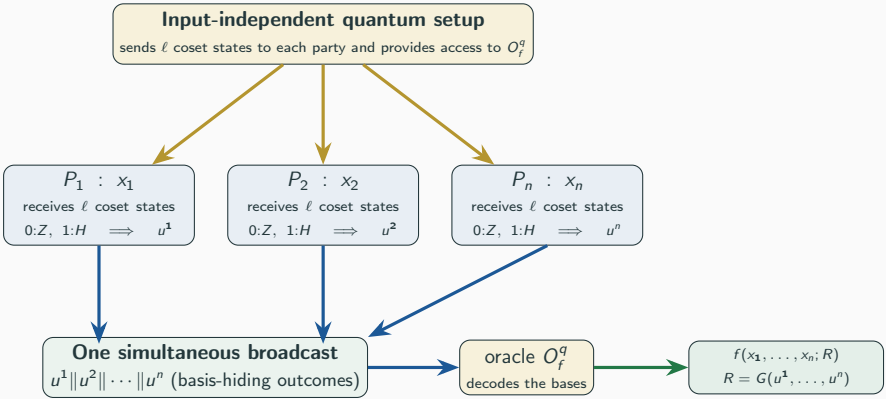
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Theorem

Information-theoretically secure one-message MPC for *all* randomized functionalities with quantum pre-processing in the oracle model. Up to $n-1$ corruptions leak *at most one output* of $f(x_H, \cdot)$.

Classical Pre-Processing: Substitution Attacks

Construction under attack

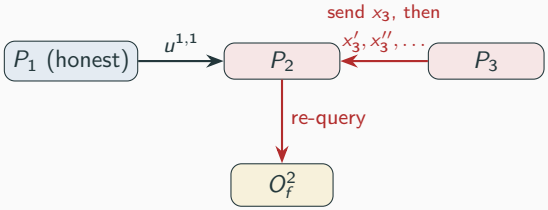
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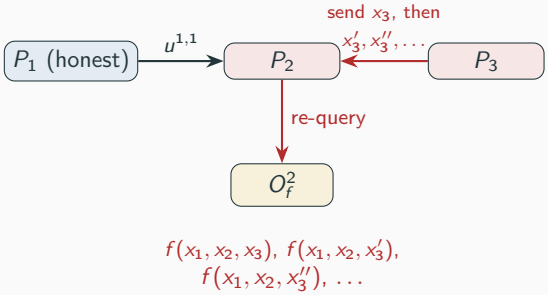
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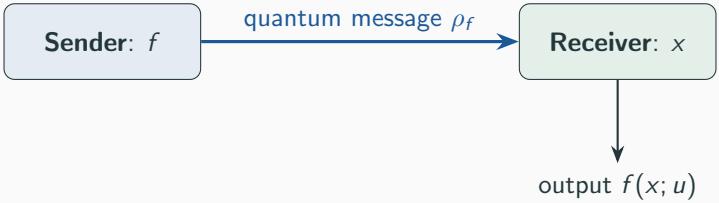
Evaluator P_2 can learn the entire residual function

$$f(x_1, x_2, \cdot).$$



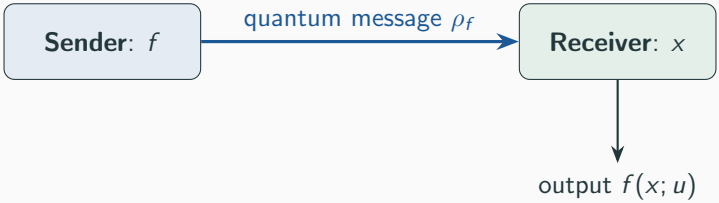
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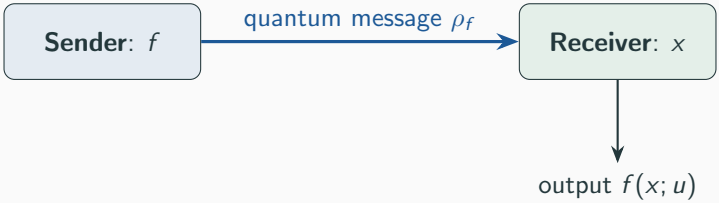


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- **vs. malicious sender:** standard simulation security.

From PCF to a QOTP: Evaluation and Consequence

QOTP package for f_k

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Consequence from LWE

PC-PRFs from LWE, together with the QOTP transformation for PCFs, yield **semi-honest one-message 2PC** for pseudorandom functionalities in the plain model from **LWE alone**.

Landscape: Two-Party Computation

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This work	plain	pseudorandom	semi-honest + operational	1 output / **LWE**

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We remove iO for semi-honest pseudorandom functionalities and obtain malicious security for the broader constrained class.

Summary

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- **2PC:** Semi-honest one-message 2PC for pseudorandom functionalities from **LWE** (plain model) and **malicious** one-message 2PC for constrained functionalities from **iO** (CRS model).

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- **2PC:** Semi-honest one-message 2PC for pseudorandom functionalities from **LWE** (plain model) and **malicious** one-message 2PC for constrained functionalities from **iO** (CRS model).

Open Questions

1. Can the MPC oracles be instantiated, yielding one-message MPC in the *plain model*?

Summary

Takeaway: quantum information shrinks adversarial leakage in one-message secure computation, from the whole residual function down to one (or c) outputs.

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Open Questions

1. Can the MPC oracles be instantiated, yielding one-message MPC in the *plain model*?
2. Can standard assumptions give plain-model 2PC beyond pseudorandom functionalities?